01 / ENGINEERING TECHNICAL NOTE

A symmetric bolted timber joint

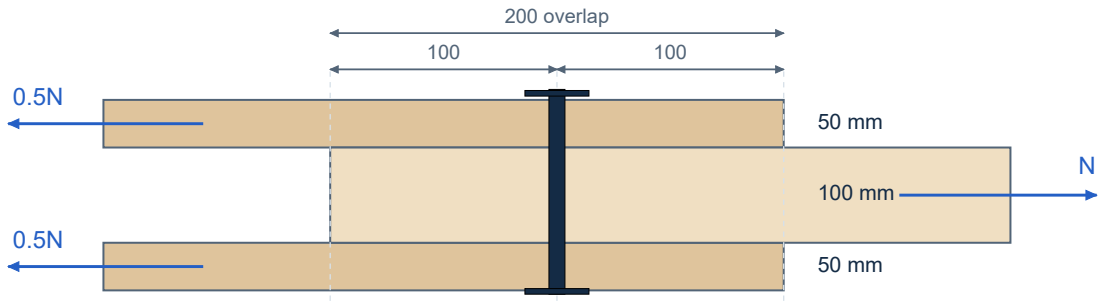
Ambient-temperature worked example

A reproducible study of force transfer, Johansen resistance, geometric detailing and time-dependent movement, with a proposed experimental programme.

10.026 kN	0.703	1.198 mm
Two-plane design resistance	Selected component utilisation	Calculated final slip

Interpretation
The selected M12 configuration satisfies the component checks implemented here. Brittle failure, bolt steel resistance, washer bearing and physical performance remain unresolved. **No overall connection design pass is established.**

SIDE ELEVATION



All timber grain parallel to the action. Balanced load introduction is assumed.
Schematic detail; dimensions govern. Bolt axis and thicknesses are exaggerated.
Figure 1. One bolt, two shear planes and one transfer station. Each outer member carries half the axial action.

Defined baseline

Item	Prescribed input
Timber	C24 solid timber; centre 100 mm; two outer members 50 mm each
Geometry	Width 200 mm; overlap 200 mm; bolt centred in the overlap and width
Fastener	M12 grade 8.8; Ø13 mm timber hole; two Ø40 × 4 mm washers
Actions	Permanent tensile action 3 kN; variable tensile action 2 kN
Model	Service class 2; parallel to grain; smooth shank at both planes; zero rope effect, friction and pretension credit

C24 reference properties: ρ_k = 350 kg/m³, ρ_m = 420 kg/m³ and $f_{t,0,k}$ = 14.5 MPa [1]. These are reference values, not verified product declarations. The selected bolt tensile strength is 800 MPa.

02 / CALCULATION MODEL

Resistance and force transfer

The full axial action passes through the central member and divides equally between the outer members. Balanced loading and identical outer members are required. Compression reversal, bending and out-of-plane forces are excluded.

$$N_{Ed} = 1.35G_k + 1.5Q_k = 7.05\ \text{kN}, \qquad N_{ser} = G_k + Q_k = 5.00\ \text{kN} \tag{1}$$

The action factors are illustrative inputs. They have not been confirmed for a project or National Annex. Resistance uses first-generation EN 1995-1-1 expressions, with source locators in [1, 3].

Material-dependent fastener properties

$$f_{h,0,k} = 0.082(1 - 0.01d)\rho_k = 25.256\ \text{MPa} \tag{2}$$

$$M_{y,Rk} = 0.3f_{u,k}d^{2.6} = 153.491\ \text{kN mm} \tag{3}$$

Here d = 12 mm, t_1 = 50 mm and t_2 = 100 mm. Identical timber gives an embedment-strength ratio β = 1. The following characteristic resistances are **per bolt and per shear plane**.

Double-shear Johansen modes

$$F_{g,Rk} = f_{h,1,k}t_1 d = 15.154\ \text{kN}, \qquad F_{h,Rk} = \frac{f_{h,2,k}t_2 d}{2} = 15.154\ \text{kN} \tag{4}$$

$$F_{j,Rk} = \frac{1.05 f_{h,1,k}t_1 d}{2+\beta}\left[\sqrt{2\beta(1+\beta) + \frac{4\beta(2+\beta)M_{y,Rk}}{f_{h,1,k}dt_1^2}} - \beta\right] = 8.146\ \text{kN} \tag{5}$$

$$F_{k,Rk} = 1.15\sqrt{\frac{2\beta}{1+\beta}}\sqrt{2M_{y,Rk}f_{h,1,k}d} = 11.092\ \text{kN} \tag{6}$$

Mode j governs at 8.146 kN per shear plane. Apply the two-plane factor once, using k_{mod} = 0.8 and γ_M = 1.3:

$$R_d = 2\frac{k_{mod}}{\gamma_M}\min(F_{g,Rk}, F_{h,Rk}, F_{j,Rk}, F_{k,Rk}) = 10.026\ \text{kN} \tag{7}$$

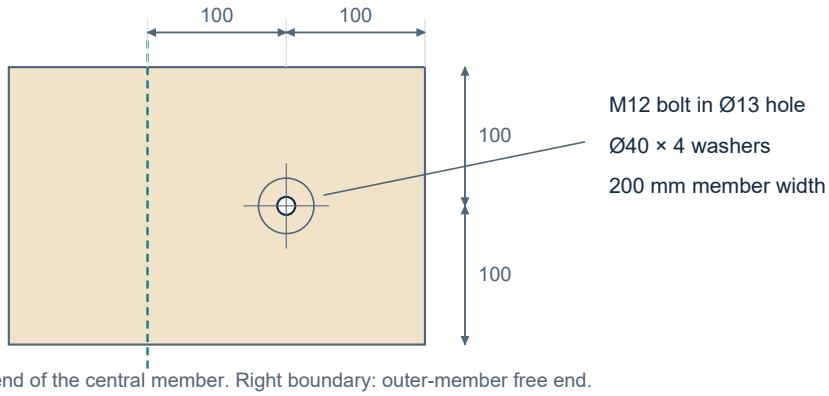
Selected component result
The 7.05 kN design action gives a utilisation of **0.703**. This covers the selected Johansen model only. Brittle timber failure, bolt steel resistance and washer bearing require separate assessment.

Sources: [1], pp.36-38 and 44; [3], 8.2.2, Eq.8.7 and 8.5.1.1, Eqs.8.30-8.33. Formulae use N and mm before conversion to the displayed units.

03 / GEOMETRY CHECKS

Detailing and nominal net tension

PLAN OF OVERLAP



Dashed line: hidden free end of the central member. Right boundary: outer-member free end.

Figure 2. Centred bolt and washer footprint. Dimensions are in mm. This is a calculation detail, not a fabrication drawing.

Geometric check	Actual / mm	Required / mm	Result
First loaded end	100.0	≥ 84.0	Satisfied
Opposite loaded end	100.0	≥ 84.0	Satisfied
Both centred edges	100.0	≥ 36.0	Satisfied
Timber hole diameter	13.0	≤ 13.0	Satisfied
Washer outer diameter	40.0	≥ 36.0	Satisfied
Washer thickness	4.0	≥ 3.6	Satisfied
Washer radius / available face	20.0	≤ 100.0	Satisfied
Washer diameter / timber hole	40.0	> 13.0	Satisfied

Inter-fastener and inter-row spacing do not apply to one bolt. The footprint checks do not establish washer bearing, splitting or end tear-out resistance. Washer-hole dimensions and complete product geometry remain unverified.

Nominal net-section tension

$$A_{net} = (b - d_0)t, \qquad R_{t,d} = A_{net}\frac{k_{mod} f_{t,0,k}}{\gamma_M} \tag{8}$$

Member	Net area / mm²	Action / kN	Resistance / kN
Centre	18,700	7.050	166.862
One outer member	9,350	3.525	83.431

Both nominal net-section utilisations are 0.0423. No favourable tensile size increase is used. This nominal stress check does not model fracture, splitting or shear-out. Sources: [1], pp.22 and 45; [3], 6.1.2, Table 8.4 and 10.4.3.

04 / SERVICEABILITY MODEL

Movement and load duration

The two identical shear-plane springs act in parallel. Mean density, ρ_m = 420 kg/m³, gives a post-seating linear stiffness. It does not describe the full measured response of a fabricated joint.

$$K_{ser,\,plane} = \frac{\rho_m^{1.5} d}{23} = 4.491 \text{ kN/mm}, \qquad K_{joint} = 2K_{ser,\,plane} \tag{9}$$

$$u_{inst} = \frac{N_{ser}}{K_{joint}} = \frac{5.00}{8.9817} = 0.5567 \text{ mm} \tag{10}$$

Action-wise creep estimate

For identical service-class-2 timbers, the selected first-generation model uses member k_{def} = 0.8 and connection factor $2k_{def}$ = 1.6. The variable-action factor ψ_2 = 0.3 is prescribed for this study and needs project confirmation [1-3].

$$u_{fin,\,calc} = \frac{G_k(1+2k_{def}) + Q_k(1+2\psi_2 k_{def})}{K_{joint}} = 1.1980 \text{ mm} \tag{11}$$

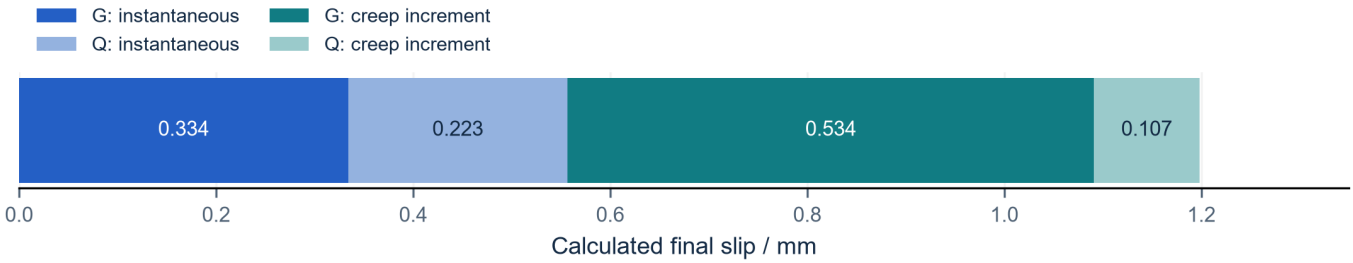


Figure 3. The calculated final slip comprises instantaneous movement and action-specific creep increments. Values are rounded in the graphic; the full-precision total is 1.1980 mm.

Free play and movement budget
The baseline additional free play is zero. A prescribed total free-play allowance is added once, without creep amplification, and is not inferred from the timber hole diameter. The **2 mm study budget is an illustrative choice**, not a code limit.

Separate load-duration cases

Action case	Design action / kN	k_{mod}	Resistance / kN	Utilisation
Combined G + Q	7.050	0.8	10.026	0.703
Permanent G only	4.050	0.6	7.520	0.539

The combined case governs this selected envelope. The instantaneous characteristic variable action remains in Eq.(11); only its creep increment is reduced by ψ_2. The envelope is not a complete set of project combinations.

Sources: [1], pp.7-8 and 32-33; [2], pp.157-159, Table 5.4 and Eq.5.86; [3], 2.2.3(5), 2.3.2.2(3) and 7.1/Table 7.1.

05 / CONTROLLED SENSITIVITY

What changes with bolt diameter?

Timber, member dimensions and model topology are fixed; hole diameter is d + 1 mm. Loaded-end distances remain 100 mm. Washer diameter is the larger of 40 mm and 3d; thickness is the larger of 4 mm and 0.3d. These are analytical scenarios, not optimisation or measured behaviour.

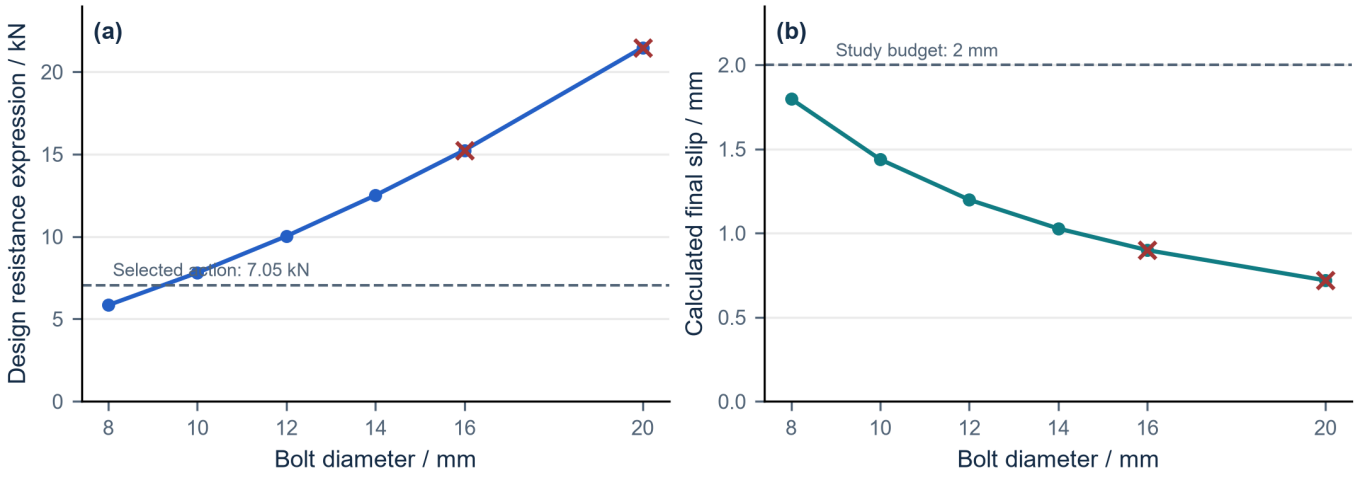


Figure 4. (a) Selected two-plane resistance expression; (b) calculated final slip with zero added free play. Red crosses mark failure of the fixed 100 mm loaded-end distance. Washers increase as prescribed above and are listed below.

Bolt / mm	Washers Ø × t / mm	Expression / kN	Utilisation	Geometry checks	Movement / mm
8	40 × 4.0	5.840	1.207	Satisfied	1.797
10	40 × 4.0	7.807	0.903	Satisfied	1.438
12	40 × 4.0	10.026	0.703	Satisfied	1.198
14	42 × 4.2	12.505	0.564	Satisfied	1.027
16	48 × 4.8	15.242	0.463	Failed	0.898
20	60 × 6.0	21.472	0.328	Failed	0.719

Interpretation
For M8, the prescribed action exceeds the selected Johansen resistance. With the prescribed larger washers, M14 satisfies the geometry checks. M16 and M20 require loaded-end distances of 112 and 140 mm, respectively, so both fail the fixed 100 mm detail. Larger Johansen resistance cannot compensate for these failures. The M12 baseline retains the unresolved mechanisms identified on pp.1-4.

Sensitivity to seating and sustained action

M12 scenario	Final movement / mm	Against 2 mm study budget
Zero added free play	1.198	Below
0.5 mm added free play	1.698	Below
1.0 mm added free play	2.198	Above
2.0 mm added free play	3.198	Above

Varying ψ_2 from 0 to 1 gives calculated final slips of 1.091 to 1.447 mm. These are prescribed inputs, not measured assembly tolerances. Source: the full-precision sensitivity JSON/CSV supplied with the case; model and limits are defined on pp.2-4.

06 / PHYSICAL VALIDATION PLAN

Proposed experimental programme

Programme status: proposed
No specimens have been fabricated or tested. The objective is to challenge the predicted mechanism, post-seating stiffness and load-sharing assumptions. The proposed nine tests do not establish a characteristic resistance or population coverage.

Series	Change from baseline	Repeats	Research question
A	M12; 50 mm outer members	3	How closely do measured slip and failure match the symmetric model?
B	M10; Ø11 mm timber hole	3	What changes when bolt diameter reduces at fixed member dimensions?
C	M12; 35 mm outer members	3	Does reduced outer thickness change the limiting mechanism?

Rig, measurements and bias control

Record	Purpose
Balanced clevis or equalising fixture	Load the centre against both outer members. The laboratory must resolve grips, restraint, capacity and guarding before manufacture.
Calibrated force measurement	Retain load-cell identification, range, calibration and force history.
Paired local slip transducers	Measure relative movement on both sides to identify joint slip, rotation and unequal outer-member response.
Actuator displacement and photographs	Separate fixture/member deformation from local slip; document seating, cracks, embedment, ovalisation and post-test bolt deformation.
As-built member and fastener records	Measure geometry, moisture, density and defects for each member. Record product certificates, fit and assembly condition.

Procedure and retained evidence

Before testing. Establish the rig and instrument ranges, record alignment, then use a pilot to choose a capacity estimate. Finalise loading history, rate and analysis windows against the licensed EN 26891 text and the approved laboratory procedure.

During and after testing. Separate seating from post-seating response. Report declared stiffness windows, peak force and observed failure mode. Compare each specimen with a calculation based on its own recorded properties. Retain raw data, scripts, photographs and anomalous tests.

EN 26891 provides the strength/deformation testing framework [5]. A later cyclic programme would require separate planning under EN 12512 [6]. Only their public scope pages were checked; complete test procedures have not been implemented.

07 / DEVELOPMENT BOUNDARIES

From joint specimen to assembly

The worked calculation is one component of a wider development process. The following work packages identify the evidence needed before the detail could support an assembly-level design decision.

Manufacture, assembly and disassembly

Stage	Required evidence
Detail coordination	Grain direction, free ends, hole alignment, shank position, head/nut access, product grade and washer geometry on coordinated drawings.
Trial fabrication	Workshop record of drilling damage, fit, washer seating and balanced assembly. Tightening must not introduce unverified friction credit.
Assembly trial	Tools, access, assembly time, tolerances and defects. Record actual condition rather than assuming nominal fit.
Disassembly and possible reuse	Hole condition, cracks, embedment damage and bolt straightness after removal. Any reuse proposal needs measured condition and appropriate tests.

An explicit specimen inventory

A proposed coupon uses one 100 × 200 × 600 mm central piece and two 50 × 200 × 600 mm outer pieces: **0.024 m³ nominal timber** before waste. The 35 mm outer-member variant uses 0.0204 m³. The 600 mm lengths are planning assumptions. Grip length and boundary-condition design remain open.

Assembly interfaces and wider performance

Topic	What remains unresolved
Environmental assessment	Product-specific environmental declarations, actual mass, transport, machining waste, fixings and a declared life-cycle boundary. No quantified saving or life-cycle assessment is claimed.
Renovation-panel integration	Insulation, sheathing, moisture management and the load path into the existing building need coordinated assessment.
Fire performance	Exposure, protection continuity and connection behaviour need a separate assembly assessment. An isolated-member section-loss illustration cannot establish joint or assembly fire resistance.
Numerical extension	A later finite-element study needs a timber material model, contact and bolt representation, mesh assessment and separate calibration/validation evidence. No FE input or solver job forms part of this case.

These are proposed development tasks. They do not represent completed fabrication, environmental analysis, fire verification or physical validation.

08 / EVIDENCE AND LIMITATIONS

Verification and source record

Reproducibility and calculation checks

The suite passed **179 calculation/export regression tests and 16 release-evidence checks (195 total)**. Of the 179 calculation tests, 62 cover the joint case. Python/browser numerical parity covers **42 vectors**, 3,960 numeric and 1,480 classification comparisons, and 12 expected rejections, with no failures. These are implementation checks within declared fixtures, not evidence of physical validity.

Full-precision inputs, results, sensitivity data and parity records accompany the calculation source. Automated checks compare the Python implementation with the browser calculation routines and check the release-evidence records. Browser interface review and deployment evidence are recorded separately.

Outstanding checks
Current applicable standard and National Annex; project actions and product declarations; bolt steel and washer bearing resistance; splitting, shear-out and block failure; fabrication and durability; rotational effects; cyclic degradation and repeated assembly; connection/assembly fire performance; and physical/long-term validation.

Sources and calculation locators

[1] Swedish Wood/TDUK. *Design of timber structures, Volume 2*, UK edition 1:2022. Properties p.10; factors pp.7-8; net tension p.22; slip pp.32-33; resistance pp.36-38; bolts and detailing pp.44-45. View source.

[2] Swedish Wood. *Design of timber structures, Volume 1*, edition 3:2022. Connection creep pp.157-159, Table 5.4 and Eq.5.86. View source.

[3] BS EN 1995-1-1:2004+A1:2008, publicly hosted copy. Used for selected first-generation clause cross-checks; this is not the current official standards service. View source.

[4] BSI. UK National Annex and Eurocode coexistence statement. Not every applicable National Annex choice has been checked; second-generation 2025 provisions are not implemented here. View source.

[5] BSI. BS EN 26891: scope of strength and deformation testing for mechanically fastened timber joints. View source.

[6] BSI. BS EN 12512: scope of cyclic testing for mechanically fastened timber joints. View source.

Project resources

Explore the calculation code, full-precision data and interactive case through the project repository.